

Code:

```
#include <stdio.h>

#define MAXN 100005

int parent[MAXN];

int find_root(int i) {
    if (parent[i] == i) return i;
    return parent[i] = find_root(parent[i]);
}

void union_nodes(int i, int j) {
    int r1 = find_root(i);
    int r2 = find_root(j);
    if (r1 != r2) {
        parent[r2] = r1;
    }
}

int rep[MAXN];

int main() {
    int n, m;
    if (scanf("%d %d", &n, &m) != 2) return 0;

    for (int i = 1; i <= n; i++) parent[i] = i;

    while (m--) {
        int a, b;
        scanf("%d %d", &a, &b);
        union_nodes(a, b);
    }

    int comp_count = 0;
    for (int i = 1; i <= n; i++) {
        if (find_root(i) == i) {
            rep[comp_count++] = i;
        }
    }
}
```

```
printf("%d\n", comp_count - 1);  
for (int i = 0; i < comp_count - 1; i++) {  
    printf("%d %d\n", rep[i], rep[i + 1]);  
}  
  
return 0;  
}
```

Input:

```
4 2  
1 2  
3 4
```

Output:

```
1  
2 4
```